JAVA
TECH COMMUNITY

DAY 5 /100

JAVA INTERVIEW - OOPS FAQ

TOP-10

ARRAY PROGRAMS

01 Write a program to sort an array in ascending order.

02 Write a program to merge two arrays into one.

03 Write a program to find the 2nd largest element in an array.

04 Write a program to remove duplicates from an array.

05 Write a program to find the missing number in an array of size N-1.

06 Write a program to find common elements between two arrays.

07 Write a program to sort 0s and 1s in an array.

08 Write a program to find the subarray with the maximum sum

09 Write a program to rotate an array by K positions.

10 Write a program to check if an array is sorted in ascending order.

1. Write a program to sort an array in ascending order.

Way 1: Simple Java

java Copy

```java
import java.util.Arrays;

public class SortArray {
    public static void main(String[] args) {
        int[] arr = {5, 2, 8, 7, 1};
        Arrays.sort(arr); // Sorting array in ascending order
        for (int num : arr) {
            System.out.print(num + " "); // Printing sorted array
        }
    }
}
```

Way 2: Java 8 (Using Streams)

java Copy

```java
import java.util.Arrays;

public class SortArrayUsingStreams {
    public static void main(String[] args) {
        int[] arr = {5, 2, 8, 7, 1};
        Arrays.stream(arr).sorted().forEach(num -> System.out.print(num + " "));
    }
}
```

2. Write a program to merge two arrays into one.

Way 1: Simple Java

java Copy

```java
import java.util.Arrays;

public class MergeArrays {
    public static void main(String[] args) {
        int[] arr1 = {1, 2, 3};
        int[] arr2 = {4, 5, 6};
        int[] merged = new int[arr1.length + arr2.length];

        System.arraycopy(arr1, 0, merged, 0, arr1.length); // Copying first array
        System.arraycopy(arr2, 0, merged, arr1.length, arr2.length); // Copying second array

        System.out.println(Arrays.toString(merged));
    }
}
```

Way 2: Java 8 (Using Streams)

java Copy

```java
import java.util.Arrays;
import java.util.stream.IntStream;

public class MergeArraysUsingStreams {
    public static void main(String[] args) {
        int[] arr1 = {1, 2, 3};
        int[] arr2 = {4, 5, 6};

        int[] merged = IntStream.concat(Arrays.stream(arr1), Arrays.stream(arr2)).toArray();
        System.out.println(Arrays.toString(merged));
    }
}
```

3. Write a program to find the 2nd largest element in an array.

Way 1: Simple Java

```java
import java.util.Arrays;

public class SecondLargest {
    public static void main(String[] args) {
        int[] arr = {5, 2, 8, 7, 1};
        Arrays.sort(arr); // Sorting the array
        System.out.println("Second Largest: " + arr[arr.length - 2]); // Accessing second last element
    }
}
```

Way 2: Java 8 (Using Streams)

```java
import java.util.Arrays;

public class SecondLargestUsingStreams {
    public static void main(String[] args) {
        int[] arr = {5, 2, 8, 7, 1};
        int secondLargest = Arrays.stream(arr).sorted().distinct().toArray()[arr.length - 2];
        System.out.println("Second Largest: " + secondLargest);
    }
}
```

4. Write a program to remove duplicates from an array.

Way 1: Simple Java

```java
import java.util.HashSet;
import java.util.Set;

public class RemoveDuplicates {
    public static void main(String[] args) {
        int[] arr = {1, 2, 2, 3, 4, 4, 5};
        Set<Integer> set = new HashSet<>();
        for (int num : arr) {
            set.add(num); // Adding unique elements to a set
        }
        System.out.println(set);
    }
}
```

Way 2: Java 8 (Using Streams)

```java
import java.util.Arrays;

public class RemoveDuplicatesUsingStreams {
    public static void main(String[] args) {
        int[] arr = {1, 2, 2, 3, 4, 4, 5};
        int[] unique = Arrays.stream(arr).distinct().toArray(); // Removing duplicates
        System.out.println(Arrays.toString(unique));
    }
}
```

5. Write a program to find the missing number in an array of size N-1.

Way 1: Simple Java

java

```java
public class FindMissingNumber {
    public static void main(String[] args) {
        int[] arr = {1, 2, 4, 5, 6};
        int n = arr.length + 1; // Total numbers (including missing one)
        int totalSum = n * (n + 1) / 2; // Sum of first n numbers
        int arraySum = 0;

        for (int num : arr) {
            arraySum += num; // Sum of given numbers
        }

        int missing = totalSum - arraySum; // Missing number
        System.out.println("Missing Number: " + missing);
    }
}
```

Way 2: Java 8 (Using Streams)

java

```java
import java.util.Arrays;

public class FindMissingNumberUsingStreams {
    public static void main(String[] args) {
        int[] arr = {1, 2, 4, 5, 6};
        int n = arr.length + 1; // Total numbers
        int totalSum = n * (n + 1) / 2; // Sum of first n numbers
        int arraySum = Arrays.stream(arr).sum(); // Sum of given numbers
        System.out.println("Missing Number: " + (totalSum - arraySum));
    }
}
```

6. Write a program to find common elements between two arrays.

Way 1: Simple Java

java

```java
import java.util.HashSet;

public class FindCommonElements {
    public static void main(String[] args) {
        int[] arr1 = {1, 2, 3, 4};
        int[] arr2 = {3, 4, 5, 6};

        HashSet<Integer> set = new HashSet<>();
        for (int num : arr1) {
            set.add(num); // Add elements of first array to set
        }

        for (int num : arr2) {
            if (set.contains(num)) { // Check if set contains element from second array
                System.out.print(num + " ");
            }
        }
    }
}
```

Way 2: Java 8 (Using Streams)

java

```java
import java.util.Arrays;
import java.util.List;
import java.util.stream.Collectors;

public class FindCommonElementsUsingStreams {
    public static void main(String[] args) {
        Integer[] arr1 = {1, 2, 3, 4};
        Integer[] arr2 = {3, 4, 5, 6};

        List<Integer> common = Arrays.stream(arr1)
            .filter(Arrays.asList(arr2)::contains)
            .collect(Collectors.toList());
        System.out.println(common);
    }
}
```

7. Write a program to sort 0s and 1s in an array.

Way 1: Simple Java

java Copy

```java
public class SortZerosOnes {
    public static void main(String[] args) {
        int[] arr = {0, 1, 1, 0, 1, 0};
        int countZero = 0;

        for (int num : arr) {
            if (num == 0) countZero++; // Count number of 0s
        }

        for (int i = 0; i < arr.length; i++) {
            arr[i] = (i < countZero) ? 0 : 1; // Fill array with 0s and 1s
        }

        for (int num : arr) {
            System.out.print(num + " ");
        }
    }
}
```

Way 2: Java 8 (Using Streams)

java Copy

```java
import java.util.Arrays;

public class SortZerosOnesUsingStreams {
    public static void main(String[] args) {
        int[] arr = {0, 1, 1, 0, 1, 0};
        arr = Arrays.stream(arr).sorted().toArray(); // Sort array
        System.out.println(Arrays.toString(arr));
    }
}
```

8. Write a program to find the subarray with the maximum sum.

Way 1: Simple Java (Kadane's Algorithm)

java Copy

```java
public class MaxSubarraySum {
    public static void main(String[] args) {
        int[] arr = {-2, 1, -3, 4, -1, 2, 1, -5, 4};
        int maxSum = Integer.MIN_VALUE, currentSum = 0;

        for (int num : arr) {
            currentSum = Math.max(num, currentSum + num);
            maxSum = Math.max(maxSum, currentSum);
        }

        System.out.println("Maximum Subarray Sum: " + maxSum);
    }
}
```

Way 2: Java 8 (Using Streams - Not efficient for this case)

java Copy

```java
import java.util.Arrays;

public class MaxSubarraySumUsingStreams {
    public static void main(String[] args) {
        int[] arr = {-2, 1, -3, 4, -1, 2, 1, -5, 4};
        int maxSum = Arrays.stream(arr).reduce(0, (a, b) -> Math.max(a + b, b)); // Kadane logic
        System.out.println("Maximum Subarray Sum: " + maxSum);
    }
}
```

9. Write a program to rotate an array by K positions.

Way 1: Simple Java

java

```java
import java.util.Arrays;

public class RotateArray {
    public static void main(String[] args) {
        int[] arr = {1, 2, 3, 4, 5};
        int k = 2; // Number of positions to rotate
        int n = arr.length;

        int[] rotated = new int[n];
        for (int i = 0; i < n; i++) {
            rotated[(i + k) % n] = arr[i]; // Rotating by K positions
        }

        System.out.println(Arrays.toString(rotated));
    }
}
```

Way 2: Java 8 (Using Streams)

java

```java
import java.util.Arrays;

public class RotateArrayUsingStreams {
    public static void main(String[] args) {
        int[] arr = {1, 2, 3, 4, 5};
        int k = 2; // Number of positions to rotate
        int[] rotated = Arrays.stream(arr)
            .boxed()
            .collect(Collectors.toList())
            .stream()
            .mapToInt(i -> arr[(arr.length - k + i) % arr.length])
            .toArray();
        System.out.println(Arrays.toString(rotated));
    }
}
```

10. Write a program to check if an array is sorted in ascending order.

Way 1: Simple Java

java

```java
public class CheckSorted {
    public static void main(String[] args) {
        int[] arr = {1, 2, 3, 4, 5};
        boolean isSorted = true;

        for (int i = 0; i < arr.length - 1; i++) {
            if (arr[i] > arr[i + 1]) {
                isSorted = false; // Break if order is violated
                break;
            }
        }

        System.out.println("Is array sorted? " + isSorted);
    }
}
```

Way 2: Java 8 (Using Streams)

java

```java
import java.util.Arrays;

public class CheckSortedUsingStreams {
    public static void main(String[] args) {
        int[] arr = {1, 2, 3, 4, 5};
        boolean isSorted = Arrays.stream(arr)
            .boxed()
            .sorted()
            .toArray(Integer[]::new)
            .equals(Arrays.stream(arr).boxed().toArray(Integer[]::new));
        System.out.println("Is array sorted? " + isSorted);
    }
}
```

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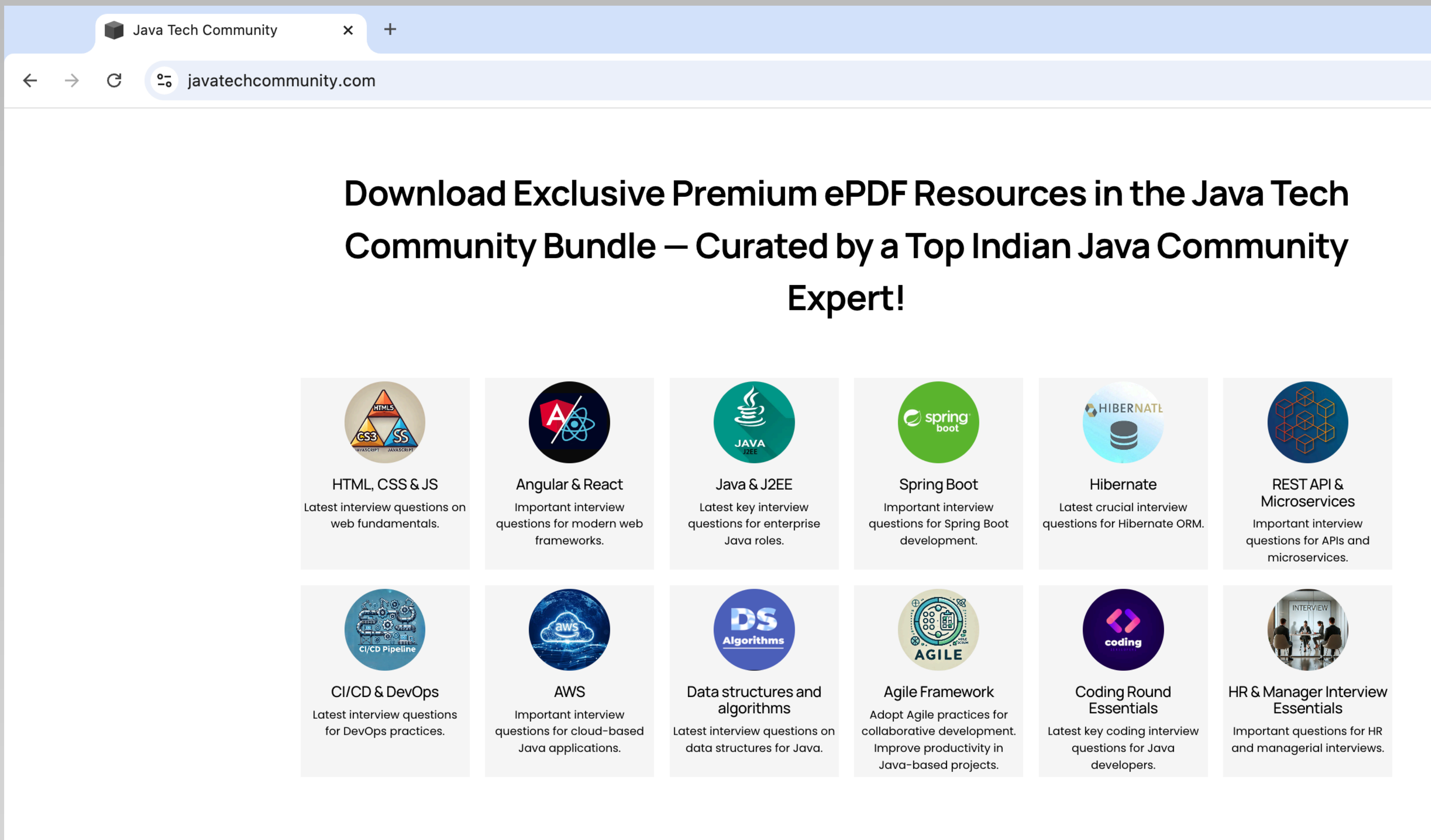

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